



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.NOS.9

# Represent Polygons on the Coordinate Plane

Lesson 7-7 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

**Language Objective:** I can explain how I found a distance using the words vertices, coordinates, and absolute value.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

vertex

polygon

perimeter

distance between vertices

scale



Tap any word to see what it means and a picture.

1

— A corner point of a polygon, named by an ordered pair on the coordinate plane.

2

— A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.

3

— The total distance around a polygon — the sum of all its side lengths.

4

— When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.

5

— What one grid unit stands for in the real situation.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **vertex** — A corner point of a polygon, named by an ordered pair on the coordinate plane.  
*vértice* — Un punto de esquina de un polígono, nombrado por un par ordenado en el plano de coordenadas.

☐ **polygon** — A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.  
*polígono* — Una figura cerrada hecha de segmentos; en el plano se dibuja marcando los vértices y conectándolos en orden.

☐ **perimeter** — The total distance around a polygon — the sum of all its side lengths.  
*perímetro* — La distancia total alrededor de un polígono: la suma de todos sus lados.

☐ **distance between vertices** — When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.  
*distancia entre vértices* — Cuando dos vértices comparten una coordenada x o y, la distancia es la diferencia absoluta de las otras coordenadas.

☐ **scale** — What one grid unit stands for in the real situation.  
*escala* — Lo que representa una unidad de la cuadrícula en la situación real.

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The park's width is \_\_\_\_ feet. Using the area formula, its length is \_\_\_\_ feet, which is \_\_\_\_ units on the map — so the new vertices sit that many units below the known ones.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** When two vertices share the same y-coordinate, the segment between them is horizontal, so its length comes from the DIFFERENCE in the coordinate that changes — the x-coordinates.

**Evidence:** A strong answer explains that the SHARED coordinate tells you the direction of the side, and the DIFFERING coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

**Reasoning:** This evidence connects the definition of polygon to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The park's width is \_\_\_\_ feet. Using the area formula, its length is \_\_\_\_ feet, which is \_\_\_\_ units on the map — so the new vertices sit that many units below the known ones.
- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- I know because \_\_\_\_.  
*Lo sé porque \_\_\_\_.*
- So \_\_\_\_.  
*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from  $y = 3$  down to  $y = -3.75$  the distance is  $3 + 3.75 = 6.75$  units, not  $3.75 - 3 = 0.75$ . Distances that cross zero are the whole reason the absolute value is there.*
7. **Try It 1:**
8. **Try It 2:**